Routing Team: Interim EM Assessment and Plan

Created 2026-07-20
Last Major Update 2026-07-28
Contact Tim Song
Status Draft

Summary

I recently stepped in as interim Engineering Manager for the Routing team for the quarter. During my first week, I met individually with every engineer on the team as well as Product Manager. These conversations gave me a picture of a team that is technically capable and deeply committed, but operating under conditions that are making it difficult to perform at its best.

This document shares my observations, identifies the highest-leverage opportunities for improvement, and proposes a path forward. The intent is not only to stabilize the Routing team this quarter, but to build processes and practices that can hopefully serve as a model for other teams.

In the spirit of continuous improvement, my goal is to continue listening and learning from the team. I will continue to identify opportunities that could add value as well as revisit any changes to make sure they continue to add value.

Culture of Team

- Shared ownership and executing more as a team

DLQ Stability

- Bring visibility and priority to DLQ stability with Product
- Focus on future system health not just clearing the queues

Large Volume of Support Requests

- Offload repeatable requests to new Support team

Engineering and Product Alignment

- Build transparency with Engineering initiatives

North Star

My goal for the Routing team is to become a high-functioning, self-sustaining engineering team that can absorb change, deliver reliably, and operate with confidence. By the end of the quarter, the team should be capable of onboard new Engineers and a new Engineering Manager without feeling overloaded. Three dimensions define what success looks like:

Team Health and Scalability

Knowledge is distributed across the team, not concentrated in one person. Onboarding is structured, documented, and repeatable. The team can absorb attrition and new headcount without a crisis in the future. By the end of the quarter, the team is in a position to hand off to a new EM with confidence.

Engineering and Product Alignment

Engineering and Product share a single prioritized backlog backed by a transparent allocation framework. Priorities are made explicitly in a joint refinement process. System reliability and engineering investment are first-class citizens alongside product work, not afterthoughts. The relationship between Engineering and Product is healthy when Product understands and advocates for engineering investment.

Operational Maturity

The team runs with clear intake, visibility, and planning processes. Work is legible to the team, leadership and stakeholders. The processes built here are documented and repeatable, and become second nature for the team.

Opportunities

Below I share the immediate opportunities for the team.

Culture of Team

What

The Routing team currently operates in a siloed fashion where individual engineers own workstreams end to end with little cross-team visibility or collaboration. While this model can feel efficient in the short term, it concentrates knowledge within individuals and leaves the rest of the team unaware of what others are working on or how systems function. Over time this creates bottlenecks, single points of failure, and a team that is fragile to change.

Current State

Attrition over the past six months has left the team with a significant knowledge gap, and the siloed culture means that gap is difficult to close. New engineers coming onto the team have limited visibility into how work gets done or how systems fit together, making onboarding slow and heavily dependent on a small number of people. Requests flow directly to the engineer perceived as the owner of a given area, bypassing any intake or prioritization process and making workloads invisible to the rest of the team and leadership.

Path Forward

The goal is a team where knowledge, ownership and responsibility are shared. No single person should be the only one who can answer a question or resolve an issue.

- Establish an intake process for all requests routed through Conner or Tim, creating a single prioritization point and preventing unplanned work from landing directly on any one engineer
- Limit the number of WIP priorities to build space to operate more as a team
- Continue to define an owner for a feature workstream, but shift to executing on features as a team to drive more collaboration and shared knowledge
- Prioritize documentation and training artifacts so the team's knowledge is codified and not dependent on any one person's availability

Longer term, the measure of success is that the team can absorb the loss of any one person without a crisis and onboard a new engineer or a new EM without the process depending on tribal knowledge.

DLQ Stability

What

The Routing team's architecture depends heavily on async message passing between services to drive functionality. When a consumer cannot process a message, it lands in a dead-letter queue (DLQ) as an exception state. Growing DLQs in Routing indicate stalled request inventory or state syncing issues between Retrieval and Platform systems. Managing these DLQs requires effort from the engineering team to diagnose the root cause of why the message failed. More discovery of the DLQ issue can be found here.

Current State

The Routing team currently manages 40+ queues that other teams in Platform and Retrieval Ops are dependent on. The volume of queues requiring triage because of the increase of DLQ messages continues to be a challenge day to day, and one that the team hasn't been able to solve. This dashboard can give you a sense of the issue. The other concern is that Product does not have the needed visibility to understand what the impact is and how much effort is required to resolve the DLQs and engineering has struggled to provide a good estimate. Product has struggled with building a predictable roadmap with this uncertainty.

Path Forward

DLQ stability is not a maintenance task. It is a system reliability investment that directly impacts customers and operational partners. Treating it as such requires both visibility and shared ownership between Engineering and Product.

- Establish DLQ resolution as a standalone initiative, separate from KTLO, to make the investment and progress visible
- Introduce Product into the triage process of DLQs so that impact, level of effort, and prioritization decisions are made jointly. If an issue is a Product gap, Product needs to own the risk of deferring it
- Under the Engineering and Product alignment framework, DLQ resolution sits in the engineering investment bucket and is protected by the 30% allocation
- Jamie was leading the effort to help resolve the errors impacting Chart Compass, could he support this initiative with Routing

Long term, the team needs to invest in the design of their message driven system and see if it still drives the outcome that is needed to support Switchboard. Or whether there is a better approach to managing its domain.

Large Volume of Support Requests

What

Routing sits at the center of all communication between Switchboard and every retrieval source. This position means the team receives a disproportionate volume of support requests from clients and ops partners. The requests are mostly for rerouting inventory from another one retrieval method to another, and updating coordinate details such as address or phone information to allow the system to rematch.

Current State

Support requests are consuming a significant portion of the team's capacity. While individual requests are not complex, the volume is high enough that Product has to batch the requests weekly. The team has no protected path to focus on roadmap and stability work.

Path Forward

The near-term solution is offloading repeatable support requests, like rerouting inventory or updating addresses on inventory, to the Support team being formed. Runbooks and utilities already exist, making Pete and the Support team a viable path with structured knowledge transfer.

We could use the Routing support requests as a pilot for how to effectively use the new Support team being proposed to build out the Support function. Long term, Product is thinking about self-service tooling built into Switchboard to help CS be able to support these requests themselves without the engineering team having to be involved.

Outlier: Member Locator Project Support Requests

Member Locator project support requests are a special case. Historical knowledge of this process is limited on the current team. Before any handoff, the team needs to document the request types and establish a formal intake process. We need to assess whether the Support team can eventually absorb these as well. Long term, we need to have conversations about the ownership for Member Locator as Routing might not be the best fit.

Engineering and Product Alignment

What

There is ongoing tension between Engineering and Product around roadmap priorities. Engineers want to invest in system improvements that reduce support requests and on-call burden. Product wants the team focused on client-facing feature work. The team currently lacks the process to have these tradeoff conversations explicitly, resulting in too many priorities in flight and constant context switching.

Current State

Product does not have clear visibility into engineering-driven work like DLQ cleanup, making it difficult to plan accurately or set expectations with stakeholders. There is no shared definition of what the team's true priorities are at any given time.

Path Forward

Alignment requires both a structural guardrail and a collaborative process.

Allocation Framework

The team will operate on a quarterly allocation target of 60% product work, 30% engineering and security investment, and 10% support and KTLO. This becomes the forcing function that prevents engineering investment from being crowded out by product or reactive work.

Joint Refinement Process

Within that allocation, Engineering and Product will run regular joint refinement sessions. Engineers own the engineering investment bucket and are responsible for surfacing proposals, articulating the problem and impact, and recommending priority. Product does the same for their bucket. The group stack ranks across both with the allocation as a guardrail keeping the balance honest.

The health of this relationship will be measured by whether Product understands and advocates for engineering investment work, not just tolerates it. Over time the goal is a team where Engineering and Product trust each other's judgment and make tradeoffs together.

Supporting Practices

- Design documents and one-pagers for significant engineering and product work, giving both sides better context before refinement
- A visible board showing all inflight work across product, engineering investment, and support, making overcommitment visible in real time
- Planning sessions each quarter to set the allocation, review the backlog, and align on commitments before they are made to stakeholders

High-Level Process:

1. Product sets up a kickoff with supporting documentation about what we are building and why (PRD)
2. Owning engineer creates one pager about what will be built and how (spike); breaks down the work at a high level and reviews with team for feedback and questions
3. Continuously plan and refine the work with the team for execution